Module 06 Lab Reflective Journal

Keira Burdine

In Module 6, I learned more about the difference between regression and classification by working with the Titanic dataset in Google Colab. I used linear regression to predict ticket fares based on passenger age and class, and logistic regression to predict whether a passenger survived based on age, passenger class, and sex. Completing both activities helped me understand that regression predicts numerical values, while classification predicts categories.
I practiced preparing the dataset, selecting features and a target, dividing the data into training and testing sets, and then fitting each model. One important lesson was understanding why we use different evaluation methods. My linear regression model produced a mean squared error of 3,364.92 and a root mean squared error of about $58.01. My logistic regression model had a test accuracy of 74.83%. I learned that the first result measures the size of prediction errors, while accuracy measures the percentage of correct classifications. I also learned to interpret the linear regression coefficients. The coefficient for age was approximately -0.44, and the coefficient for passenger class was approximately -38.30. These values described how the model predicted ticket fares would change when one feature increased while the other stayed the same.
One challenge I faced was making small typing mistakes in Python, including misspelling method and model names. Reading the error messages and correcting my code helped me become more comfortable troubleshooting instead of starting over. I also worked through a Markdown formatting issue in my Knowledge Check responses. Overall, this lab gave me more confidence in using Colab, understanding machine-learning model outputs, and paying attention to details in my code.

John Grubbs

I enjoyed this assignment because it helped me better understand the difference between regression and classification. Using the Titanic dataset showed me that regression predicts a number, like a passenger's fare, while classification predicts a category, like whether someone survived.

I also learned that different models need different evaluation methods. Mean squared error worked for predicting fares, while accuracy worked better for predicting survival. Looking at the regression coefficients helped me understand how features like age and passenger class affected the predictions.

Overall, this assignment gave me practice cleaning data, training models, making predictions, and checking how well the models worked.

Luis Adrian

In this lab, I learned the difference between regression and classification by using the Titanic dataset. I used linear regression to predict a passenger's fare based on their age and passenger class. Then, I used logistic regression to predict whether a passenger survived based on their age, class, and sex. The examples helped me understand that regression predicts numerical values, while classification predicts categories.

One important part of this lab was learning how to evaluate each model. My linear regression model had a **Mean Squared Error of 3364.92** and a **Root Mean Squared Error of 58.01.** My logistic regression model achieved an accuracy of **74.83%.** Understanding MSE was more challenging because it measures squared errors instead of a percentage of correct answers. I also learned that the coefficients show how each feature changes the predicted fare while the other feature stays the same.